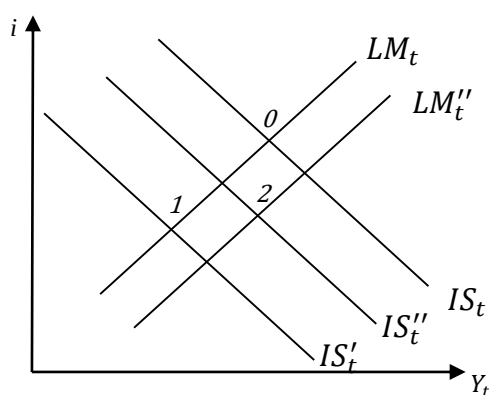


Question 1

- a) In country Alpha, a closed economy with constant prices, in period $t + 1$ (“the future”) consumption depends only on disposable income at $t + 1$, and investment on the values that the interest rate and output take on at $t + 1$ only. Turning now to the current (t , “the present”), consumption demand is increasing in individuals’ total wealth, $C_t = C[\text{total wealth}]$, while investment demand is increasing in current and expected future income levels, and decreasing in the current and expected future levels of interest rates. Finally, M, G and T are exogenous in both periods. The initial equilibrium is shown by point ‘0’ in the graph below. At time t , Alpha experiences a stock market crash – a sudden, unexpected fall in stock prices. Show in the graph how this change will affect the time t equilibrium levels of output, interest rate and the composition of aggregate demand and discuss in details all channels through which these effects occur. (Hint: When answering neglect any eventual effect of variation in current interest rate on individual’s human wealth.)



The stock market crash causes a decrease in individuals’ financial wealth, and therefore in their total wealth. In the current period, consumption falls, the IS curve shifts to the left (IS'_t) and the equilibrium becomes point 1 in the graph. Income and current interest

rate decrease. The first one because of the reduction in the stock prices, the second because lower income leads to lower money demand. Since, the money supply is given there is an excess of money supply in the financial market. As for the composition of aggregate demand, government spending, G , is unchanged. consumption, C , will be lower (both because of the fall in financial wealth and the decrease in human wealth caused by the fall in current income) and the change in investment is uncertain because there is a decrease in the interest rate but also a decrease in current output, Y_t .

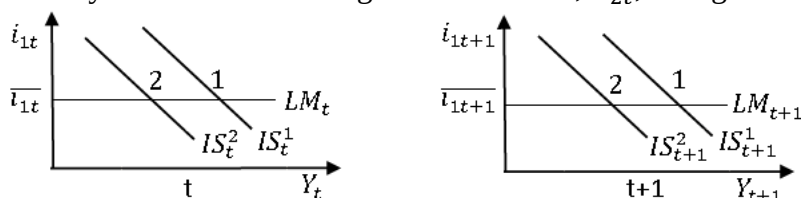
- b) Suppose the economy is described by a non-standard IS-LM model. Alpha’s central bank decides to intervene at time t (and at time t only) in order to limit the fall in stock prices you have just considered. To achieve its goal, which kind of monetary intervention (contractionary? expansionary?) should the central bank implement? Show in the graph the effects of the intervention you are suggesting assuming it is successful, so that stock prices fall less than they would have done, absent any central bank intervention and discuss the direction of the interest rate change due to the stock market crash and the central bank intervention. (Hint: Compare the final level of interest rate with the level of the interest rate at the initial equilibrium ‘0’).

The central bank should implement an expansionary monetary policy. In fact, by lowering the current interest rate, a monetary expansion will limit the fall in stock prices and raise the present discounted value of expected future dividends. Following this policy intervention, the LM curve will shift downwards (LM''_t in the graph), while the IS curve will shift to the left, but less than before, let’s say up to IS''_t in the graph (remember that by assumption the central bank intervention leads to a fall in the stock prices smaller than the one that would have occurred otherwise). The new equilibrium point is ‘2’ in the graph, where the change in output has an uncertain sign. However, the level of the interest rate is certainly lower than the one prevailing at the initial equilibrium (‘0’).

Question 2

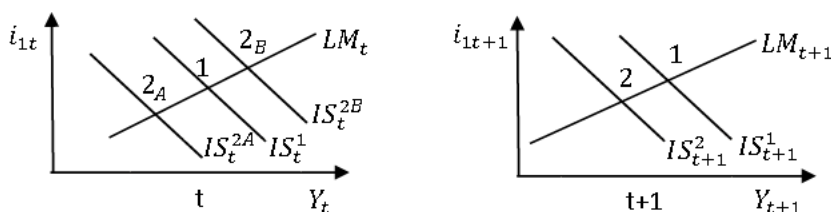
In country Macrolandia, there are only one-year and two-year bonds. The current and the expected future inflation rates are both zero, so that the real and the nominal interest rates are equal. In period $t + 1$ (“the future”), a standard IS-LM model describes the functioning of the economy. In particular, in the year $t + 1$ consumption depends only on disposable income at $t + 1$, and investment on the interest rate and output at $t + 1$ only. Turning now to the current period (t , “the present”), consumption is increasing in both current and expected future disposable income, $C_t = C(Y_t - \bar{T}_t, Y_{t+1}^e - \bar{T}_{t+1}^e)$, investment depends positively on current and expected future income levels, and negatively on the current and expected future levels of short-term interest rates, the demand for money has the usual functional form, and G and T are exogenous as usual. Starting from an initial equilibrium, at time t the government of Macrolandia announces a decrease of its purchases of goods and services at time $t + 1$, $\Delta \bar{G}_{t+1} < 0$.

- a. Assuming that, in both periods, Macrolandia’s central bank chooses the interest rate, describe and represent in the two graphs below the effects of this fiscal policy announcement on income and the interest rate levels prevailing at time t and at time $t + 1$. How will the time t yield to maturity on the two-year bonds circulating in Macrolandia, i_{2t} , change? Why?



At time $t+1$, the cut in G will shift the IS curve to the left, thus leading to a lower equilibrium income. At time t , the ensuing decrease in expected future income will lower consumption and investment, causing a leftward shift in the IS curve. It follows that production will fall both periods, and that – given the assumption about the way the central bank behaves – both the current and the future expected short-term rates will remain unchanged. Being (approximately) equal to the average of i_{1t} and i_{1t+1}^e , and that these rates remain constant, i_{2t} will remain unchanged, too.

- b. Repeat the same analysis carried out answering the previous point assuming that, rather than the interest rate, the central bank of the country chooses the money supply. Represent in the graphs below the effects of the same fiscal policy announcement considered before and explain how will i_{2t} change in this case, and why.

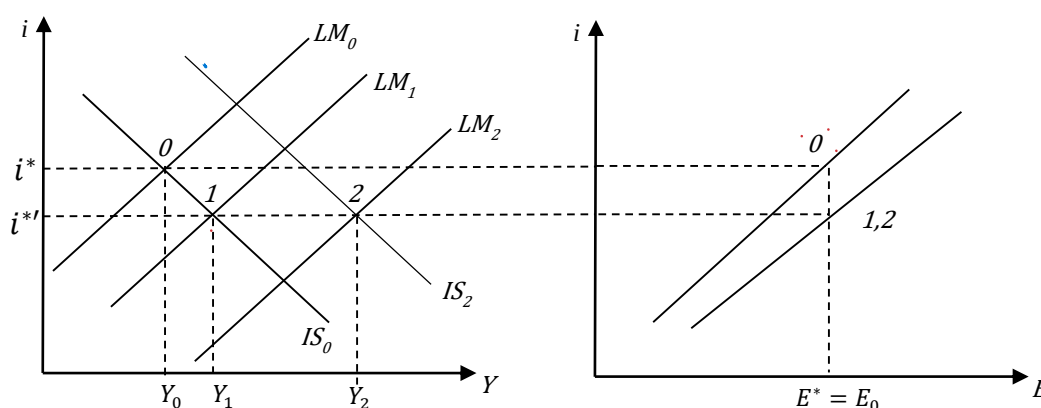


The LM curve is now positively sloped in both periods. In this case, it is not possible to conclude how, and if, i_{2t} will change. In fact, the decrease in G announced for time $t+1$ will lead to a leftward shift of the IS for that period that will now cause a fall not just in income, but also in the future interest rate (the decrease in income will lower money demand; since now the central bank keeps M constant at the chosen level, the interest rate for which the money market is in equilibrium goes down). At time t , the expectation of a lower future income decreases consumption and investment, shifting the IS to the left; at the same time, the expectation of lower future interest rate raises investment, thus tending to shift the IS to the right. The IS will shift to the left if to prevail is the first of these two opposing forces, while it will stay put if they exactly offset each other. In both cases, with a lower future expected short rate and a current short rate lower or unchanged, i_{2t} will fall. However, if to prevail is the second of the two forces mentioned above, the current short rate will go up and, given that the future expected short rate has gone down, i_{2t} could increase, decrease, or remain unchanged.

Question 3

Consider an economy that trades goods, services and financial assets with the rest of the world, under a fixed exchange rate regime. Domestic and foreign prices are constant and, for simplicity, both equal to 1 ($P = P^* = 1$). In the pair of graphs ‘open economy non-standard IS-LM’-‘interest parity condition’, show the initial equilibrium, denoting by i_0, Y_0 and E_0 the values of the interest rate, output and exchange rate at that equilibrium. Financial market participants expect that the exchange rate will be kept at fixed at the current level E_0 also in the future, so that $E^e = E_0$.

- a) Suppose that there is a fall in the foreign interest rate i^* , while foreign output Y^* and all the other exogenous variables remain constant. Assuming that the country’s central bank keeps fixing the exchange rate at E_0 , show in the graph, and explain carefully, the effects of this exogenous decrease in i^* on equilibrium output, interest rate, money supply and net exports of the country, denoting by ‘1’ the new equilibrium that will be reached.



In the right panel, the fall in the foreign interest rate leads to a downward shift in the line representing the interest parity condition; the new line will now go through – as it has to by construction – the point ($E = E^e, i = i^*$), where i^* is the new level of the foreign interest rate. Absent any intervention by the domestic central bank, the exchange rate would appreciate (it would rise above E_0). To keep it at E_0 , the central bank will have to intervene. This intervention will have to lower the domestic interest rate by the same amount by which the foreign one has gone down, thus shifting the LM curve downwards till LM_1 . In the new equilibrium (point 1), the decrease in the domestic interest rate has raised aggregate demand and equilibrium output. As for net exports, they will now be lower, since domestic income is higher and exchange rate unchanged.

- b) How would your answer have changed if, when i^* falls, foreign output Y^* had also gone up, rather than remaining constant? In the same graph used before, denote by ‘2’ the equilibrium that the economy would have reached in this case (lower i^* and higher Y^*) and compare it to the equilibrium (‘1’) where the economy settles when to vary is just the foreign interest rate. Explain.

In this case, the interest parity condition line will shift downwards, while the increase in Y^* , that – other things the same – leads to higher net exports leads to a shift of the IS curve to the right in the first panel (IS_2). Without any intervention the rightward shift of the IS would lead to an increase in domestic interest rate. Since the foreign interest rate decreases, the exchange rate would appreciate more than in the case in which the only change is due to the foreign interest rate, as in point a). Hence, the increase in domestic output is now larger, since in this case the demand for domestically produced goods goes up not only because of the fall in the interest rate, but also because of the rise in the demand by foreigners. The central bank’s intervention, aimed at keeping E constant, must

therefore be larger than before. More specifically, the domestic money supply will have to be raised by more than when the only change taking place was the one in i^* . The LM will shift to the right up to LM_2 . Now, money supply will have to go up not only to make sure that the domestic interest rate falls by the same amount as the foreign one, but also to prevent the increase in income, and therefore in money demand, caused by the rise in Y^* . The effect on NX is uncertain because both Y and Y^* increase.

Question 4

- a. Define the concepts of government deficit (in real terms), primary deficit, and public debt. Finally, write down the government budget constraint.

The government deficit in real terms is the difference between government outlays and government revenues, both measured in real terms, in a given time period (t). The stock of government (or public) debt is the sum of all the deficits incurred until time t. Finally, time t government deficit is the sum of the primary (that is, net of interest payments) deficit $G_t - T_t$ - where G is government spending on goods and services, and T denotes net taxes - and of (real) interest payments due on the stock of debt outstanding at t, rB_{t-1} :

$$Dis_t = G_t - T_t + rB_{t-1}$$

If monetary financing of the deficit is ruled out, the government budget constraint is:

$$B_t - B_{t-1} = G_t - T_t + rB_{t-1} \quad (*)$$

- b. Suppose that, until time $t-1$, the government budget was always balanced, and that government debt was equal to zero. Government spending on goods and services G and net taxes T were constant, and both equal to 20, in all periods. At time T, and only in that period, the government cuts T from 20 to 10, leaving G unchanged. At times $t+1$ and $t+2$, the government sets $G = T = 20$ once again. The deficit incurred at t because of the decision of cutting taxes in that time period is financed by issuing bonds paying a constant real interest rate equal to 5%. The government intends to repay the whole debt issued between time t and time $t+2$ at time $t+3$ (so that, from $t+3$ onwards, the stock of government bonds outstanding will be once again nil). Compute the primary surplus that the government should run at $t+3$ to achieve its goal.

Moving B_{t-1} to the right-hand side of equation () and using the quantitative information provided by the question, one computes:*

$$B_t = B_{t-1} + G_t - T_t + rB_{t-1} = 0 + 20 - 10 + 0 = 10$$

$$B_{t+1} = B_t + G_{t+1} - T_{t+1} + rB_t = 10 + 20 - 20 + 0.05 * 10 = 10.5$$

$$B_{t+2} = B_{t+1} + G_{t+2} - T_{t+2} + rB_{t+1} = 10.5 + 20 - 20 + 0.05 * 10.5 = 11.025$$

Since the government wants the stock of debt to be once again zero from $t+3$ onwards, in that period the following condition must hold:

$$0 = B_{t+2} + G_{t+3} - T_{t+3} + rB_{t+2} = 11.025 + G_{t+3} - T_{t+3} + 0.05 * 11.025$$

Solving, the primary surplus consistent with the government's objective is

$$T_{t+3} - G_{t+3} \cong 11.576.$$

[illegible]